

## Markscheme

Mathematics: analysis and approaches

Practice question bank

---

138. (a)

The complex number  $z = x + yi$  satisfies  $\arg(z + 2) = \frac{\pi}{6}$  and  $\arg(z - 2) = \frac{2\pi}{3}$ .

Find  $z$  in the form  $a + bi$ , correct to 3 significant figures.

$$\arg(z + 2) = \frac{\pi}{6}: \frac{y}{x + 2} = \tan \frac{\pi}{6} \Rightarrow y = \frac{x + 2}{\sqrt{3}} \quad (M1) \text{ A1}$$

$$\arg(z - 2) = \frac{2\pi}{3}: \frac{y}{x - 2} = \tan \frac{2\pi}{3} \Rightarrow y = -\sqrt{3}(x - 2) \quad (M1) \text{ A1}$$

$$\text{solve simultaneously: } \frac{x + 2}{\sqrt{3}} = -\sqrt{3}(x - 2) \Rightarrow x = 1 \quad (M1)$$

$$y = \sqrt{3} \approx 1.73 \quad \text{A1}$$

[7 marks]